

Asadullah Bin Rahman

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Research Interests: Image Processing, Machine Learning, Quantum Computing

Employment

- 07/25 – Present ♦ **Lecturer, Computer Science and Engineering**
World University of Bangladesh (WUB), Dhaka, Bangladesh
Roles: Undergraduate teaching, thesis & project supervision, competitive programming club mentoring, student counseling
- 07/23 – 07/25 ♦ **Research Assistant**
IoThink Lab, Department of Computer Science and Engineering, HSTU
Focus: Wavelet-based denoising and deep learning for medical image analysis; interdisciplinary dataset collection
Supervisor: Dr. Md. Abdulla Al Mamun 🔗

Education

- 07/23 – 07/25 ♦ **MSc in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
Thesis: MRI Image Denoising using Wavelet Transform and Deep Learning Architectures
CGPA: 3.547/4.00 🔗
- 01/17 – 12/22 ♦ **BSc (Engineering) in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
Thesis: MRI Brain Tumor Classification from Contrast-Enhanced MRI using Deep Learning
CGPA: 3.31/4.00 🔗

Skills

- Quantum Computing ♦ Proficient in Qiskit, Cirq, PennyLane, Classiq; applied experience in Quantum Algorithms, Quantum Machine Learning, Quantum Key Distribution
- Machine Learning ♦ Proficient in NumPy, Pandas, SciPy, Matplotlib, OpenCV, PyTorch, TensorFlow; applied experience in Image Processing, Computer Vision
- Programming ♦ C++, Java, Python; Data Structures, Algorithms, Databases; Solved 300+ problems (CodeForces, HackerRank, UVa)
- Languages ♦ Bangla (Native), English (IELTS Academic: **6.5**, CEFR Level: B2) 🔗
- Others ♦ Linux, Git, LaTeX

Research Publications

Conference Proceedings

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, “Mitigating Noise from Biomedical Images Using Wavelet Transform Techniques,” in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025, pp. 1–6. 🔗 DOI: [10.1109/ECCE64574.2025.11013062](https://doi.org/10.1109/ECCE64574.2025.11013062).
2. **A. B. Rahman**, M. Touhid Islam, M. R. Islam, M. Sohrawordi, and M. N. Sultan, “Enhanced Brain Tumor Classification from MRI Images Using Deep Learning Model,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 2023, pp. 1–6. 🔗 DOI: [10.1109/ICCIT60459.2023.10441064](https://doi.org/10.1109/ICCIT60459.2023.10441064).

Preprints

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Deep Learning Architectures for Medical Image Denoising: A Comparative Study of CNN-DAE, CADTra, and DCMIEDNet*, 2025. arXiv: [2508.17223](https://arxiv.org/abs/2508.17223) [eess.IV].  URL: <https://arxiv.org/abs/2508.17223>.
2. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Systematic Evaluation of Wavelet-Based Denoising for MRI Brain Images: Optimal Configurations and Performance Benchmarks*, 2025. arXiv: [2508.15011](https://arxiv.org/abs/2508.15011) [eess.IV].  URL: <https://arxiv.org/abs/2508.15011>.

Teaching

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| Fall 2026 | <ul style="list-style-type: none">◇ Object Oriented Programming (Batch 71C, 72D, 71E, Semester 4). CSE 06132120.◇ Introduction to Programming (Batch 74D, Semester 2). CSE 06131209. |
| Summer 2026 | <ul style="list-style-type: none">◇ Compiler Design (Batch 63A, Semester 11). CSE 06134156.◇ Object Oriented Programming (Batch 70E, Semester 4). CSE 06132120.◇ Introduction to Programming (Batch 73B, Semester 1). CSE 06131209. |
| Spring 2026 | <ul style="list-style-type: none">◇ Object Oriented Programming (Batch 69A, Semester 4). CSE 06132120.◇ Introduction to Programming (Batch 70B, 71C, Semester 2). CSE 06131209. |
| Fall 2025 | <ul style="list-style-type: none">◇ Computer Fundamentals (Batch 71G, Semester 1). CSE 06111101.◇ Object Oriented Programming (Batch 68A, Semester 4). CSE 06132120.◇ Computer and Cyber Security (Batch 63B, Semester 9). CSE 06124158. |

Thesis Supervision

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| Summer 2026 | <ul style="list-style-type: none">◇ Explainable Deep Learning for Plant Disease Classification Using Convolutional Neural Networks @ WUB: Comparison of CNN architectures for leaf disease classification on PlantVillage crops, with Grad-CAM saliency exposing the leaf regions behind each prediction.◇ ClassSync Pro: IoT-Based Smart Classroom Automation System @ WUB: LMS-scheduled control of classroom air-conditioning, fans, lighting and projectors with sensor-based occupancy validation; projected 30–40% cut in campus energy use. |
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Coaching and Mentoring

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| 09/26 – Present | <ul style="list-style-type: none">◇ Competitive Programming Coach @ WUB: Training undergraduate teams for ICPC (Asia West Continent / Dhaka Regional), NCPC and inter-university programming contests (IUPC). Weekly problem-solving sessions and mock contests covering data structures, graph algorithms, dynamic programming, number theory and computational geometry; team formation, contest strategy and post-contest editorial review. |
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Research and Projects

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| 02/26 | <ul style="list-style-type: none">◇ Multilayered Security : Implemented and simulated a research project on Multi-Layered Security System: Integrating Quantum Key Distribution with Classical Cryptography to Enhance Steganographic Security. |
| 11/24 | <ul style="list-style-type: none">◇ Guava Fruit Disease Dataset @ IoThink Lab : Collaborated on dataset collection for interdisciplinary research. |

Research and Projects (continued)

- o8/24 ♦ **Quantum Machine Learning Projects @ Womanium Program**: Four QML projects completed for *QML for Conspicuity Detection in Production*.
 - **QML for Anomaly Detection** 🔗: hybrid quantum-classical model for anomaly detection.
 - **Quantvolutional Neural Networks** 🔗: hybrid quantum convolutional model for MNIST digit classification.
 - **Quantum Variational Classifier** 🔗: quantum classifier for penguin species classification.
 - **Quantum Regression Model** 🔗: learned and predicted the sine function on the interval $[0, 2\pi]$.

Professional Development

- o9/24 – 12/24 ♦ **QClass24/25 Fall Semester** 🔗: Achieved 96% in 3 ECTS graduate-level program on Quantum Algorithms and QKD.
- o6/24 – o8/24 ♦ **Womanium Quantum + AI 2024** 🔗: Coursework on Quantum Computing and AI.
- o8/24 ♦ **QGSS 2024 – Quantum Excellence** 🔗 🔗: Intensive quantum computing boot camp by IBM Qiskit.
- ♦ **Classiq Diploma** 🔗: Advanced Quantum Algorithm design.
- ♦ **PennyLane Diploma** 🔗: Quantum Machine Learning Challenge completion.
- o7/24 ♦ **QNickel Diploma** 🔗: Quantum Algorithms workshop.
- ♦ **QBronze Diploma** 🔗: Introductory Quantum Computing and Programming.
- o9/20 ♦ **Python for Everybody Specialization** 🔗: Coursera specialization on Python fundamentals and data structures.

Awards and Achievements

- o6/25 ♦ **Unitary Hack 2025** 🔗 🔗: For open source contributions to the quantum software ecosystem.
- o4/25 ♦ **YQuantum 2025** 🔗 🔗: Solved BlueQubit's Peaked Circuits Challenge.
- o2/25 ♦ **MIT iQuHACK 2025** 🔗: Solved IonQ's Max Cut problem.
- o8/24 ♦ **Womanium Quantum + AI 2024** 🔗: Program finalist and QSL fellowship nominee.
- o6/24 ♦ **IBM Quantum Challenge 2024** 🔗 🔗: Ranked 11th globally.

Volunteering Work

- o4/25 – Present ♦ **Coordinator & Mentor, QBangladesh (QWorld)** 🔗: Mentoring students and promoting quantum computing literacy in Bangladesh.
Hosted Events: QBronze 🔗, QSilver 🔗.

Hobbies

- ♦ Chess 🔗
- ♦ Gardening